

THEORY OF COMPUTATION**(2015 Course) (Semester - I)****Time : 1 Hour]****[Max. Marks : 30****Instructions to the candidates:**

- 1) Attempt questions Q1 or Q2, Q3 or Q4, and Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

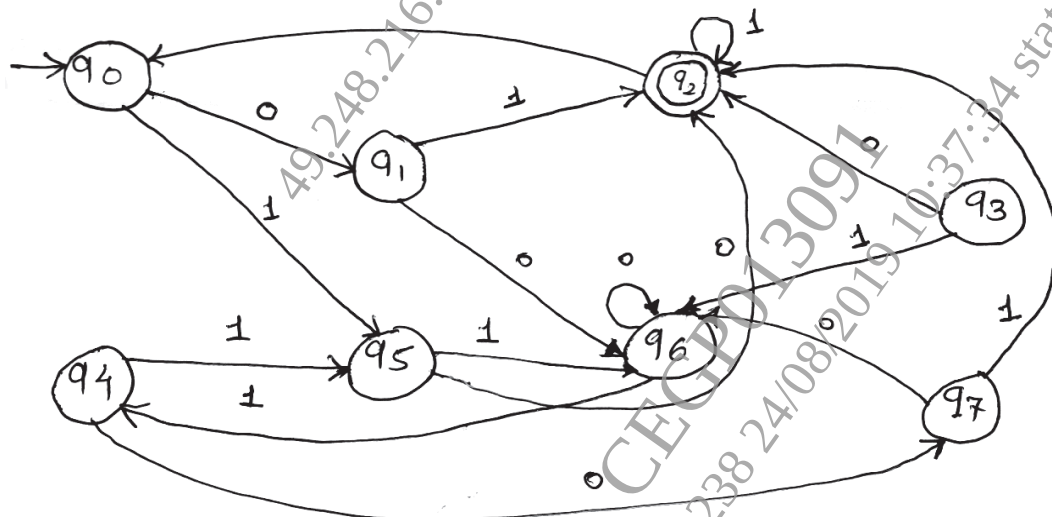
Q1) a) Construct Mealy machine to find 2' complement of any binary number & convert it into Moore machine. **[8]**

b) Define following:- **[2]**

- i) ϵ -closure of a state
- ii) NFA

OR

Q2) a) Construct the minimum state automation equivalent to the transition diagram given below. **[8]**



b) State what do you mean by FSM & state limitations of FSM. **[2]**

P.T.O.

Q3) a) Determine the regular expression over the $\Sigma = \{a, b\}$ for the following [6]

- i) Set of all strings containing exactly 2 a's
- ii) Set of all strings containing at least 2 a's
- iii) Set of all strings that do not consist of two consecutive 0's

b) Convert the given Right-linear Grammar into its equivalent Left-linear Grammar. [4]

$S \rightarrow 0A \mid 1B$

$A \rightarrow 0C \mid 1A \mid 0$

$B \rightarrow 1B \mid 1A \mid 1$

$C \rightarrow 0 \mid 0A$

OR

Q4) a) Define Pumping Lemma & Apply it to prove the following [6]

$L = \{0^i / i \text{ is an integer, } i \geq 1\}$ is not regular. [6]

b) Describe in simple english the language defined by the following regular expressions:- [4]

i) $(a+b)^* \cdot aa \cdot (a+b)^*$

ii) $a+b^* \cdot c + \epsilon$

Q5) a) Check whether or not the following Grammar is ambiguous; if it is remove the ambiguity and write an equivalent unambiguous grammar. [6]

$S \rightarrow i C t S \mid i c t S \epsilon S \mid a$

$C \rightarrow b$

b) Write note on: Chomsky Hierarchy. [4]

OR

Q6) a) Simplify the following Grammar [6]

i) $S \rightarrow aA \mid bS \mid \epsilon$

$A \rightarrow aA \mid bB \mid \epsilon$

$B \rightarrow aA \mid bc \mid \epsilon$

$C \rightarrow aC \mid bc$

ii) $S \rightarrow A \mid bb$

$A \rightarrow B \mid b$

$B \rightarrow S \mid a$

b) Find the CNF for the given CFG $S \rightarrow 0S1S \mid 1S0S \mid \epsilon$ [4]

